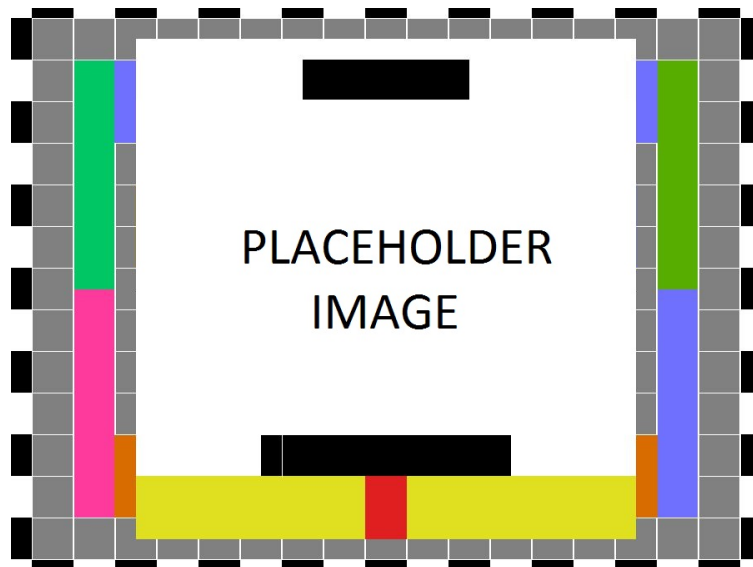


DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON

ARCHITECTURAL DESIGN SPECIFICATION
CSE 4316: SENIOR DESIGN I
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TEAM PERCEPTRON
DIGITDESK HANDWRITTEN FORM DIGITIZATION
SERVICE

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REVISION HISTORY

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1 INTRODUCTION

Architecture of DigitDesk Handwritten Form Digitization Service as an ISO/IEC/IEEE 42010:2022 [1] architecture description using IEEE 1016-2009 [2] viewpoints. Refines the SRS. Scope: the on-prem service, its REST API, and the review UI.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

Stakeholder	Primary concerns
Sponsor (records office)	accuracy, auditability, on-prem data handling, cost
Clerks (reviewers)	fast, clear review workflow
IT administrator	deployability, monitoring, security
Development team	feasibility, modular/replaceable model, testability

3 ARCHITECTURE VIEWPOINTS

Context (Section 4), Composition/Data Flow (Section 5), Logical/Functional (Sections 6–8), Deployment (Section 8). Style: a containerized pipeline of services communicating over an internal message queue and HTTP, deployed on-prem. Interfaces use #NN; subsystems named [Layer] . [Name] .

4 SYSTEM OVERVIEW & CONTEXT VIEW

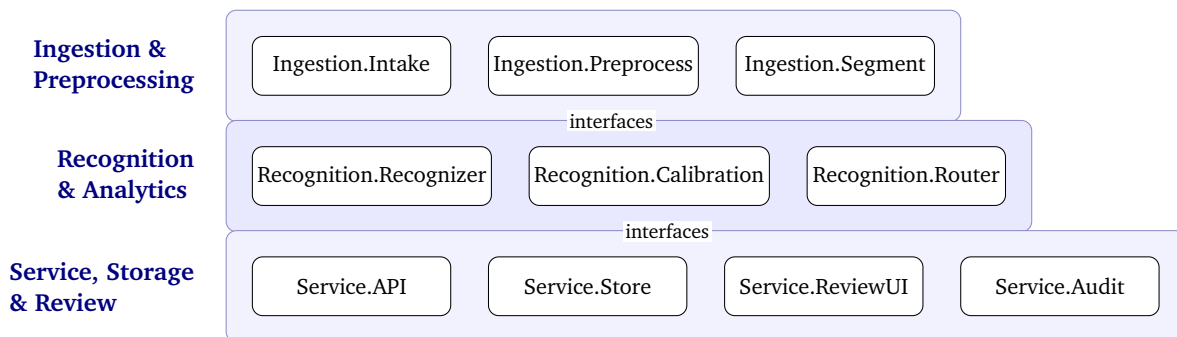


Figure 1: Three-layer pipeline: ingestion, recognition, service

External actors: the scan workflow (submits images to the API), the clerk (review UI), and the downstream records database (consumes verified export). Ingestion ingests and prepares images; Recognition recognizes and routes; Service serves, stores, reviews, and audits.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

6 INGESTION & PREPROCESSING LAYER

6.1 INGESTION.INTAKE

Responsibilities: accept uploads (EI-01), assign a job id, enqueue work.

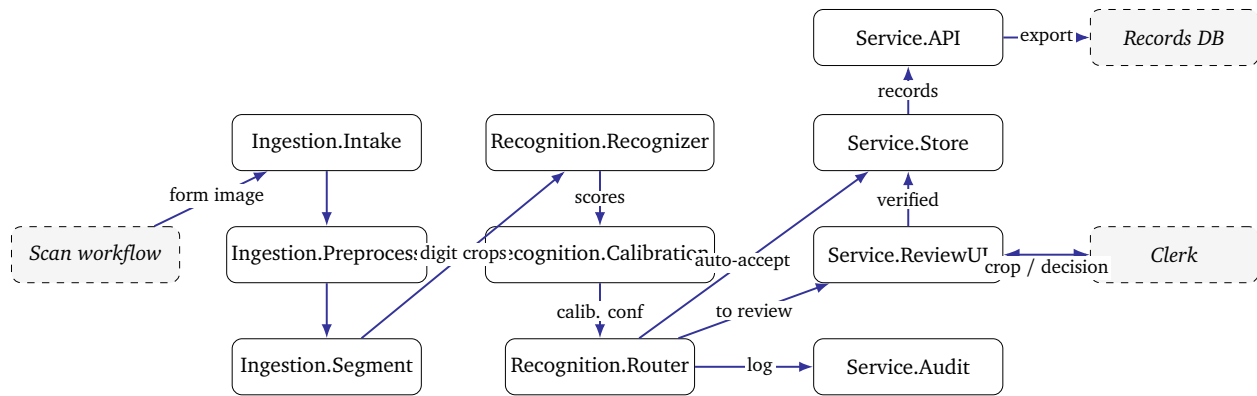


Figure 2: Image → preprocess → segment → recognize → route → store/review

ID	Description	Inputs	Outputs
#01	Ingestion API	form image	job id, queued image

6.2 INGESTION.PREPROCESS

Responsibilities: deskew, denoise, binarize to a normalized page.

6.3 INGESTION.SEGMENT

Responsibilities: locate numeric fields on the known template and emit per-field digit crops (FR-01).

Rationale: template-anchored segmentation is robust and simple for a fixed form (AD-01).

7 RECOGNITION & ANALYTICS LAYER

7.1 RECOGNITION.RECOGNIZER

Responsibilities: classify digit crops and output values with raw scores (FR-02). Model is a replaceable artifact (QA-01).

ID	Description	Inputs	Outputs
#02	Recognition	digit crops	values, raw scores

7.2 RECOGNITION.CALIBRATION

Responsibilities: convert raw scores to calibrated confidences (temperature scaling), so routing thresholds are meaningful (supports PE-02, FR-03).

7.3 RECOGNITION.ROUTER

Responsibilities: apply the acceptance threshold (FR-03) — auto-accept or enqueue for review.

8 SERVICE, STORAGE & REVIEW LAYER

8.1 SERVICE.API

Responsibilities: REST surface, authentication (SE-02), job status, export (FR-05). **Deployment:** containers orchestrated on the on-prem host.

8.2 SERVICE.STORE

Responsibilities: persist images, results, and versions (MS-01); on-prem only (SE-01).

8.3 SERVICE.REVIEWUI

Responsibilities: present queued fields for confirm/correct (FR-04).

8.4 SERVICE.AUDIT

Responsibilities: append-only log of every field decision (FR-05, SE-02).

9 ARCHITECTURE DECISIONS & RATIONALE

9.1 AD-01: TEMPLATE-ANCHORED SEGMENTATION

Context: fixed form layout (FR-01). **Decision:** anchor fields to the template rather than learn detection. **Alternatives:** learned detector (more data, more risk). **Consequences:** simple and robust now, less general (see FI-01). **Affects:** FR-01.

9.2 AD-02: CALIBRATED CONFIDENCE FOR ROUTING

Context: FR-03/PE-02 depend on trustworthy confidence. **Decision:** add a calibration stage. **Alternatives:** raw softmax (poorly calibrated). **Consequences:** extra stage, but auto-accept error stays bounded. **Affects:** FR-02, FR-03, PE-02.

10 REQUIREMENTS TRACEABILITY

Req ID	Requirement (abbrev.)	Realizing subsystem(s)
FR-01	Field segmentation	Ingestion.Segment
FR-02	Recognition + confidence	Recognition.Recognizer, Recognition.Calibration
FR-03	Confidence routing	Recognition.Router
FR-04	Human review	Service.ReviewUI
FR-05	Export & audit	Service.API, Service.Audit
SE-01	On-prem data	Service.Store

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.